Section C - TEAM 1

DIGITAL IMAGE PROCESSING ENGINE

TEAM MEMBERS

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Introduction

A zero dependency digital image processing engine.

Features:

1. Custom 2D convolution
2. Gaussian/Median blurring
3. Laplacian edge detection
4. Unsharp masking

Concepts

Custom 2D convolution

1. Rotate the kernel 180° horizontally and vertically.
2. Take scalar product of the kernel values with the neighbourhood of each pixel. Put the sum of product s value as the resultant pixel intensity.

Concepts

Median Blurring

1. Take the 3*3 neighbour pixel intensities.
2. Sort the intensities.
3. The middle one will be the resultant pixel intensity.
4. It removes the noise, and it preserves edges also.

Concepts

Gaussian Blurring

1. Given sigma and kernel size.
2. Form kernel using Gaussian blurring formula.
3. Convolve the image with the kernel.
4. If kernel size is k, normally, it takes O(K ^ 2) time per pixel.
5. But, G(x, y) = G(x, 0) * G(0, y).
6. So, we can first convolve using horizontal kernel, then vertical kernel, reducing the time taken per pixel to O(k).

Concepts

Laplacian Edge detection

1. First smoothen the image using Gaussian blurring to remove noise.
2. Kernel = [[0, 1, 0], [1, -4, 1], [0, 1, 0]]
3. Formula: d2/dx2 + d2/dy2
4. This can be used to detect the horizontal and vertical edges.
5. To detect diagonal edges also, use Kernel = [[1, 1, 1], [1, -8, 1], [1, 1, 1]].

Concepts

Unsharp masking

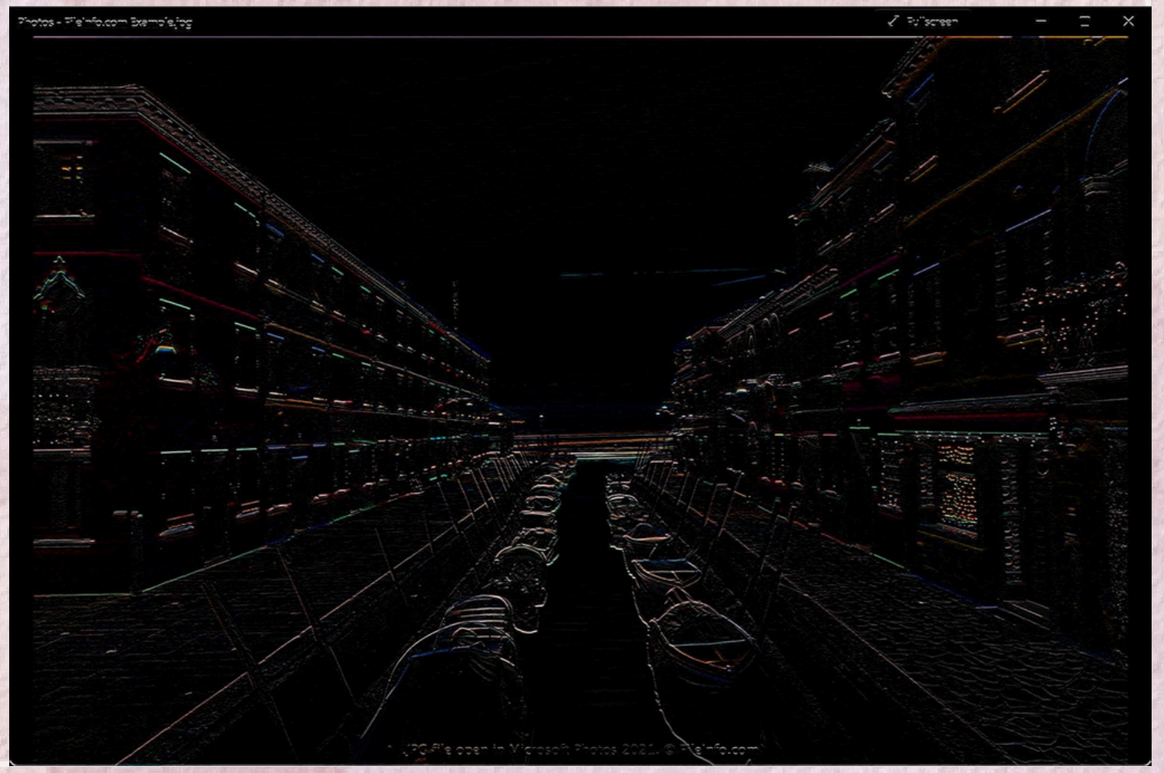
1. First smoothen the image. This will remove edges to some extent.
2. Let the smoothened image be I’(x, y).
3. Given a sharpening factor, K.
4. Resultant image will be I_res(x, y) = I(x, y) + K * (I(x, y) - I’(x, y)).

Results

Custom Convolution

Horizontal edge detection kernel used





Results

Median Blurring

Kernel size: 9



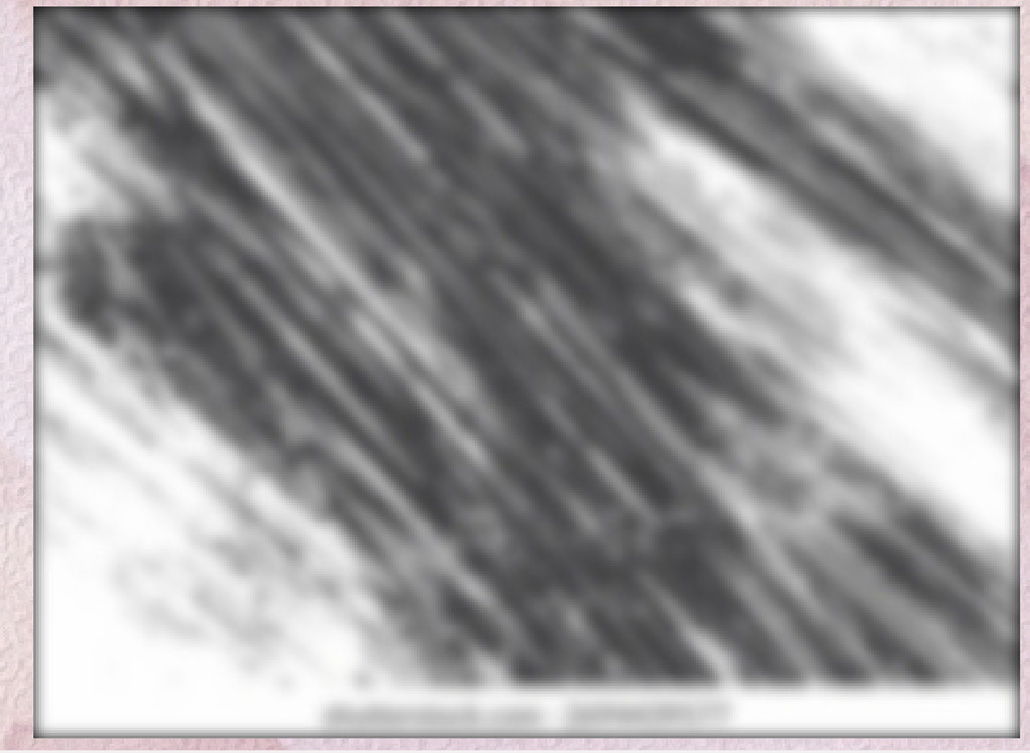


Results

Gaussian Blurring

sigma = 3, kernel size = 19





Results

Laplacian Edge detection





Results

Unsharp masking

Sharpening factor = 10





Benefits

1. No image processing library functions (like cv2) i are used except for opening and showing image. So, you can customise the features through base code.
2. Your data (images) is not going to LLM, so no data leakage problem. Image processing is done using algorithms.

Future improvements

1. Can add some more features.
2. On high resolution images, it is taking 3-4 minutes. Can find some procedure to reduce the time.

References

1. Class lecture notes for features understanding.
2. Streamlit tutorial for UI.

Github Repo Link:
https://github.com/PrakharGarg7/C_Team_1

THANK YOU